Compass-and-Straightedge Constructions

A construction uses only a compass and a straightedge — no rulers, no protractors, no measuring. Every construction in this packet is justified by a theorem from Chapter 3.

Construction 1 – A Line Through P Parallel to Line m

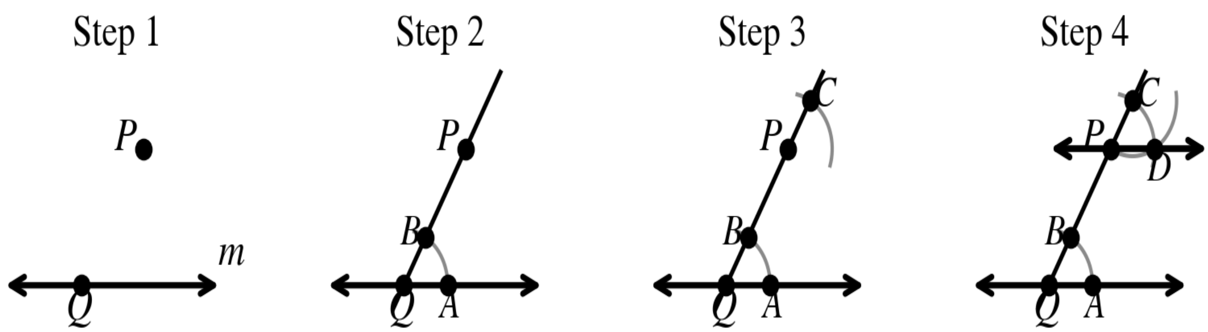
Given line m and a point P not on m, construct the line through P that is parallel to m.

Step 1. Draw line m and label a point P not on m. Choose any point Q on line m and draw line QP.

Step 2. Place the compass at Q. Draw an arc that crosses line m at A and line QP at B.

Step 3. Keeping the same compass setting, place the compass at P and draw an arc that crosses line QP at C.

Step 4. Open the compass to the distance AB. From C, draw an arc that crosses the arc from Step 3 at D. Draw line PD. This line is parallel to m.



Why it works: Steps 2–4 copy the angle at Q to the same position at P. That makes $\angle DPQ$ and $\angle AQP$ a pair of congruent corresponding angles, so $PD \parallel m$ by the Corresponding Angles Converse.

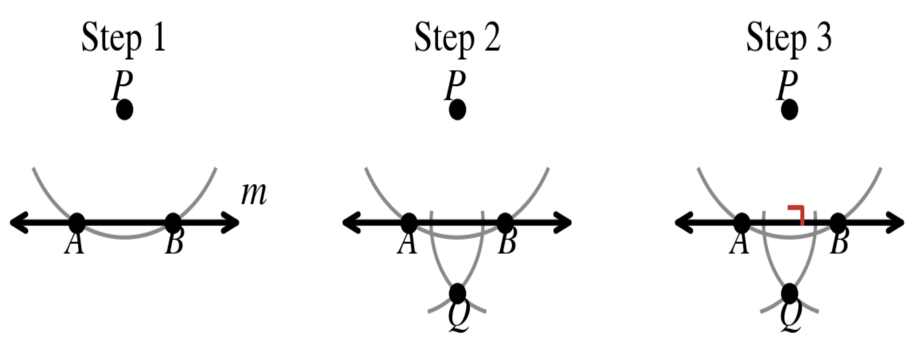
Construction 2 – A Perpendicular from a Point NOT on the Line

Given line m and a point P not on m, construct the line through P perpendicular to m.

Step 1. Place the compass at P. Open it wide enough that the arc crosses line m twice. Label the intersections A and B.

Step 2. Place the compass at A and draw an arc below line m. Keeping the same setting, place the compass at B and draw a second arc. Label the intersection of the two arcs Q.

Step 3. Draw line PQ. This line is perpendicular to line m.



Why it works: P and Q are each the same distance from A and from B, so both lie on the perpendicular bisector of $\overline{AB}$ — and that line is perpendicular to m.

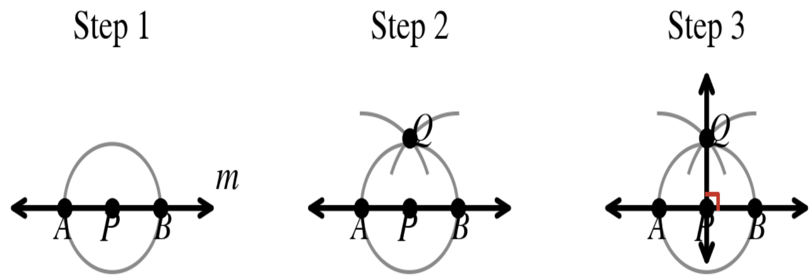
Construction 3 – A Perpendicular from a Point ON the Line

Given line m and a point P on m, construct the line through P perpendicular to m.

Step 1. Place the compass at P and draw an arc that crosses line m on both sides. Label the intersections A and B.

Step 2. Open the compass wider than PA. From A, draw an arc above the line; from B, draw a second arc with the same setting. Label their intersection Q.

Step 3. Draw line PQ. This line is perpendicular to line m at P.



Why it works: PA = PB by Step 1 and QA = QB by Step 2, so P and Q both lie on the perpendicular bisector of $\overline{AB}$.

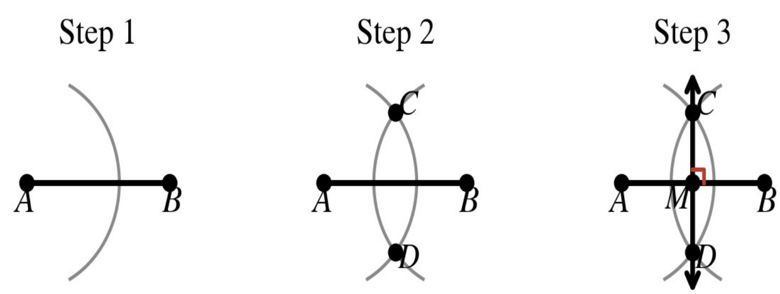
Construction 4 – The Perpendicular Bisector of a Segment

Given segment $\overline{AB}$, construct its perpendicular bisector and locate the midpoint M.

Step 1. Place the compass at A and open it to MORE than half of AB. Draw an arc above and below the segment.

Step 2. Keeping the same compass setting, place the compass at B and draw a second pair of arcs. Label the two intersections C and D.

Step 3. Draw line CD. It is perpendicular to $\overline{AB}$ and it passes through the midpoint M, so AM = MB.



Why it works: C and D are each equidistant from A and B. Two points that are equidistant from the endpoints determine the perpendicular bisector of the segment.

Name: ___________________________________

Now You Try

Use a compass and straightedge on a separate sheet. Show all construction arcs — do not erase them.

1. Draw a line j and a point R above it. Construct the line through R parallel to j. Check it: a pair of corresponding angles should measure the same.

2. Draw a segment about 4 inches long. Construct its perpendicular bisector. Both halves should measure the same, and the angle at the midpoint should be 90°.

3. Draw an acute triangle. Construct the perpendicular from one vertex to the opposite side. This segment is called an **altitude**.

4. Challenge: use Construction 3 twice to construct a square. Start with a segment $\overline{AB}$, then construct perpendiculars at A and at B.

Work space — show your arcs.